IBM Fusion

Hosted Control Planes Lab Guide

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IBM Fusion: TechZone Quick Start

Introduction

IBM Technology Zone (TechZone) is the single destination for our go-to-market teams and IBM business partners ecosystem to access on-demand and live environments to learn, build, show, and share the value of IBM solutions. Additionally, they can extend our certified base images and further customize them for test, education, demonstration, and pilot activities.

This lab guide is a Quick Start for all Fusion labs on TechZone. The lab guide will explain how to find Fusion content, as well as how to reserve and use the Fusion related lab instances. Lab exercises and specific preparation steps are detailed in the respective lab guides by topic of interest.

This guide can be used to bring a TechZone instance to a general starting point. From this starting point you may continue to the relevant lab guide document or, if you wish, create your own specific configuration for a user defined test or demo.

All TechZone instances for Fusion are based on configurable OpenShift clusters with a virtual machine (VM) node infrastructure deployed on OpenShift Virtualization (OCP-Virt) technology. The underlying VMs have nested virtualization enabled so that they can host a second level VM for demonstrating technologies such as OCP-Virt and Hosted Control Planes (HCP aka, "Control Tower").

Product disclaimer

IBM Fusion is being developed and released in an agile manner. In addition to adding new capabilities, the interface is likely to change over time. Therefore, the screenshots used in this document and other lab guides may not always look exactly like what you see in the product. Depending timing, you can expect to encounter some of the following:

- Changes in the user interface (UI), such as the location of buttons or text in various fields
- Additional tabs or buttons

These differences should not affect how the lab exercises work. When changes are observed, have patience and explore.

You will need an IBM account and corresponding IBMid to gain access to IBM Technology Zone.

Getting help

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If you require assistance in interpreting any of the steps in this demo, please post your questions to the #storage_demo_feedback Slack channel for IBMers only. Business Partners can request help at the Partner Plus Support website.

If you have an issue with TechZone or want to open a support case go to the IBM Technology Zone Help. Alternatively, you can send an email to techzone.help@ibm.com. This contact information is also provided in the emails that you receive from IBM Technology Zone.

For help with the Fusion product see the Fusion documentation.

Prerequisites & getting started

You **must** be familiar with the Linux operating system, as well as know how to use the command line interface to execute basic Linux commands such as **kubectl**, **oc**, **exit**, **ls**, **mkdir**, **ssh**, **sudo**, and **curl** (to name few). Ideally, you should also have some experience working with the Red Hat OpenShift Graphical User Interface (GUI) and executing Red Hat OpenShift Container Platform and Kubernetes command line interface (CLI) commands. If you need experience using Linux, Kubernetes and Red Hat OpenShift Container Platform, it is highly recommended that you complete the following Red Hat courses *before* you attempt to work through the exercises in this lab.

Red Hat Courses:

- DO080 Containers Kubernetes and Red Hat OpenShift Technical Overview
- Red Hat OpenShift: Technical Overview

Note Red Hat Partner Connect Training access required.

When instructions are given to enter text, the text is usually shown in mono space font `this is an example`. This is to help indicate that you need to type or copy/paste all the text as is.

IBM Fusion on TechZone

IBM TechZone can be found at this URL https://techzone.ibm.com/. To access IBM TechZone you must have an IBM ID which you can obtain from your IBM representative as a client, Business Partner or employee.

Once you are logged in to IBM TechZone head over to the IBM Fusion Collection and explore the Environement and Resources.

Bookmark these pages for future use or mark them as Favorites by clicking the heart-shaped icon on the

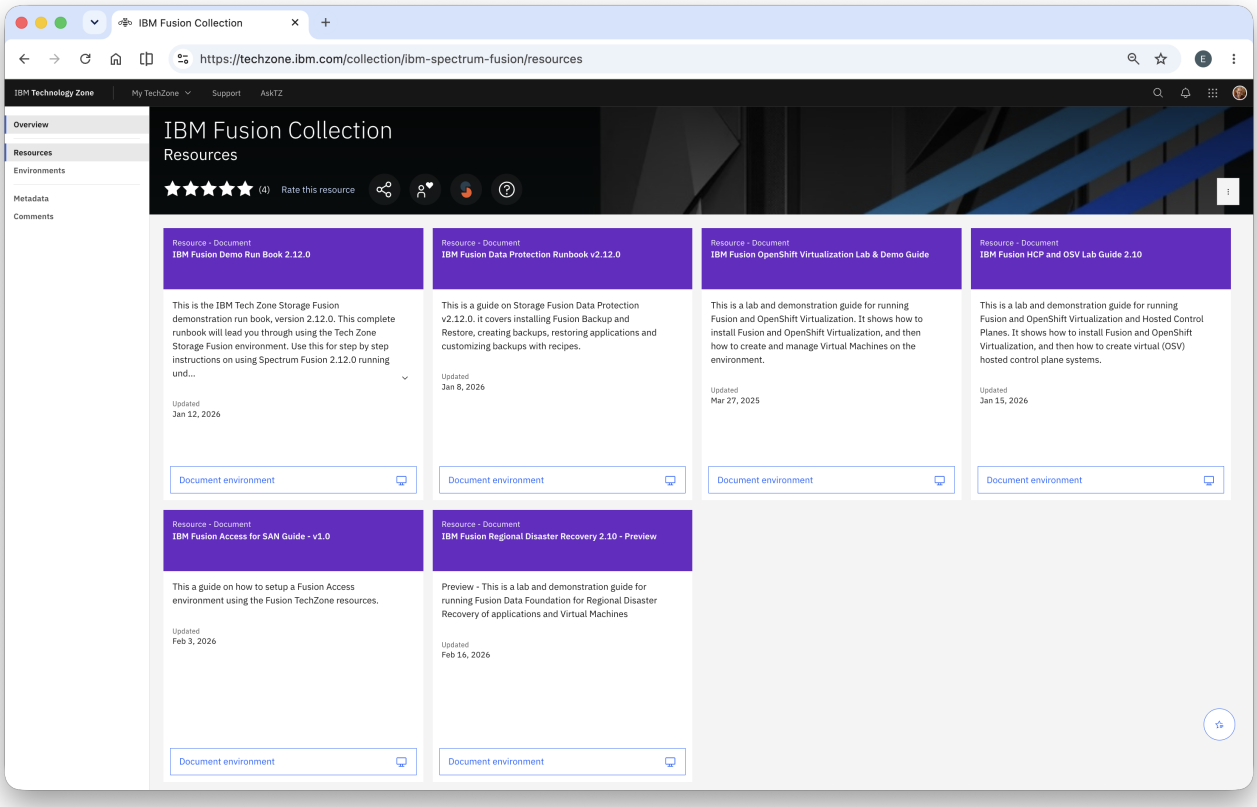
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page.

Resources

The resource page includes links to the available lab guides. These guides are available in PDF format and over time will be also available as on-line viewable material. Due to the rapid development cycles of some products, the guides may not always reflect the latest UI changes and you can expect to encounter some differences. These differences should not affect how the lab exercises work; be patience and explore to enhance your learning.

The lab guides are typically a step by step procedure to performing and learning a set of features and product capabilities. Feel free to explore and experiment!



Environments

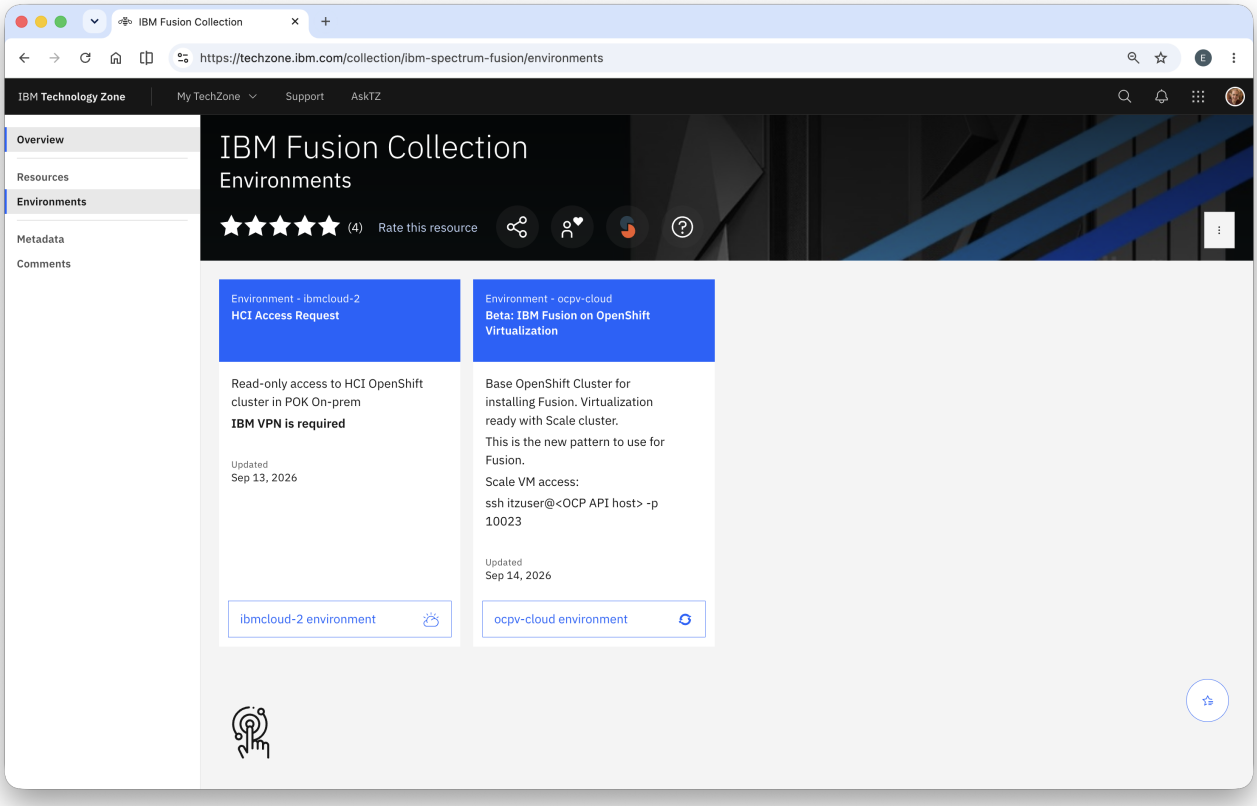
The environments page includes links to available lab resources that you can request for your project. Once your request is provisioned and ready, you will be able to use these resources for your own education, experimentation, and client demos.

The environment that is provisioned for you is an OpenShift (OCP) cluster consisting of your choice of

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components with some extra resources such as a Bastion node and an IBM Storage Scale node. You can use these for the guided labs or use them as a blank slate for your own explorations.

The next sections will walk you through a provisioning and basic use of an OCP cluster on TechZone.



Provisioning a lab

Begin your reservation process by logging in to TechZone and navigating the the Fusion Environments page.

The environments page lists the environment types you can provision and gain access to for your project.

Select the environment titled **Beta: IBM Fusion on OpenShift Virtualization** by clicking Reserve at the bottom of the tile. This will start a provisioning wizard that will walk you through the provisioning process:

1. General

Name your environment with a descriptive name that will help you identify the setup. This name will be visible to all authorized users, including team members, stakeholders, and clients.

Enter a free text description of the environment. Anyone you share this request with will see this

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description when getting access. Include relevant details to help them understand the purpose and expectations.

2. Business purpose

Describe the usage intent, including the context and purpose behind the request. This information helps the TechZone understand the business need and ensure the request is appropriately supported and prioritized.

By choosing a Purpose tile you can indicate the intended use. Each selection has a different reservation time schedule and different requirements for providing an ISC record reference. If you need to provide an ISC record reference simply copy and paste the 18-character Opportunity Number and make sure that the product list in the opportunity include the product name of the lab you are reserving; in our case Fusion.

3. Deployment

TechZone resource reservations rely on IBM Cloud. Choose a location closest to where the session will be hosted (if in person) to optimize performance. While this preference will be considered, deployment to the selected geography is subject to resource availability and cannot be guaranteed.

4. Scheduling

The scheduling page gives you the ability the set the preferred reservation date range within the Policy period determined by your intended use selection you made in the Business purpose page. Provide details on scheduling requirements and the planned dates for the intended request.

Important	Do not finalize dates with clients until the request is approved or until the resources are fully deployed and ready. Always reserve a few days in advance of any meetings or demos.

5. Configuration

Enter clear and detailed configuration choices for the environment request. You can choose to keep the defaults as displayed, or tailor the setup to your needs by clicking Customize and then modifying the environment attributes and then clicking Save.

6. Review and submit

Review the details on this pager before submitting your reservation request. This final step ensures that all required information is accurate and that you are aware of the responsibilities associated with the use of

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IBM TechZone resources.

You can go back and change reservation inputs by clicking the Edit links on the page or the Back button to step backwards in the wizard.

Any wizard step with a blue checkmark icon next to it is properly entered, and any red exclamation point icon should be revisited and corrected before submission.

Once you submit your request the system automatically start a background process to reserve the resources needed for configuring the OpenShift cluster and other resources.

You can track the reservation requests and their status on the My requests page. This page also has a display and sort filter that you can use the manage what you see on the page.

Tip Clear the filter or add the selection of Failed reservations so that you will be able to track reservations that failed to provision properly. This will prevent them from "disappearing" from your view.

You will receive emails from IBM Technology Zone <noreply@techzone.ibm.com> indicating the status of your reservation. The status may indicate a reservation failure or success, and over the duration of use you may also receive reminders for imminent expiration or confirmation of your initiated deletion.

If you receive and expiration notice and you have a date extension option, then follow up and extend your reservation in order to keep using it without disruption.

My reservation failed, what to do?

Reservations may fail from time to time. This depends of many factors including current workloads and available resources. Always reserve in advance of any meetings or demos! if a reservation fails redo the reservation entry using the reservation wizard and described in this section.

Managing a lab

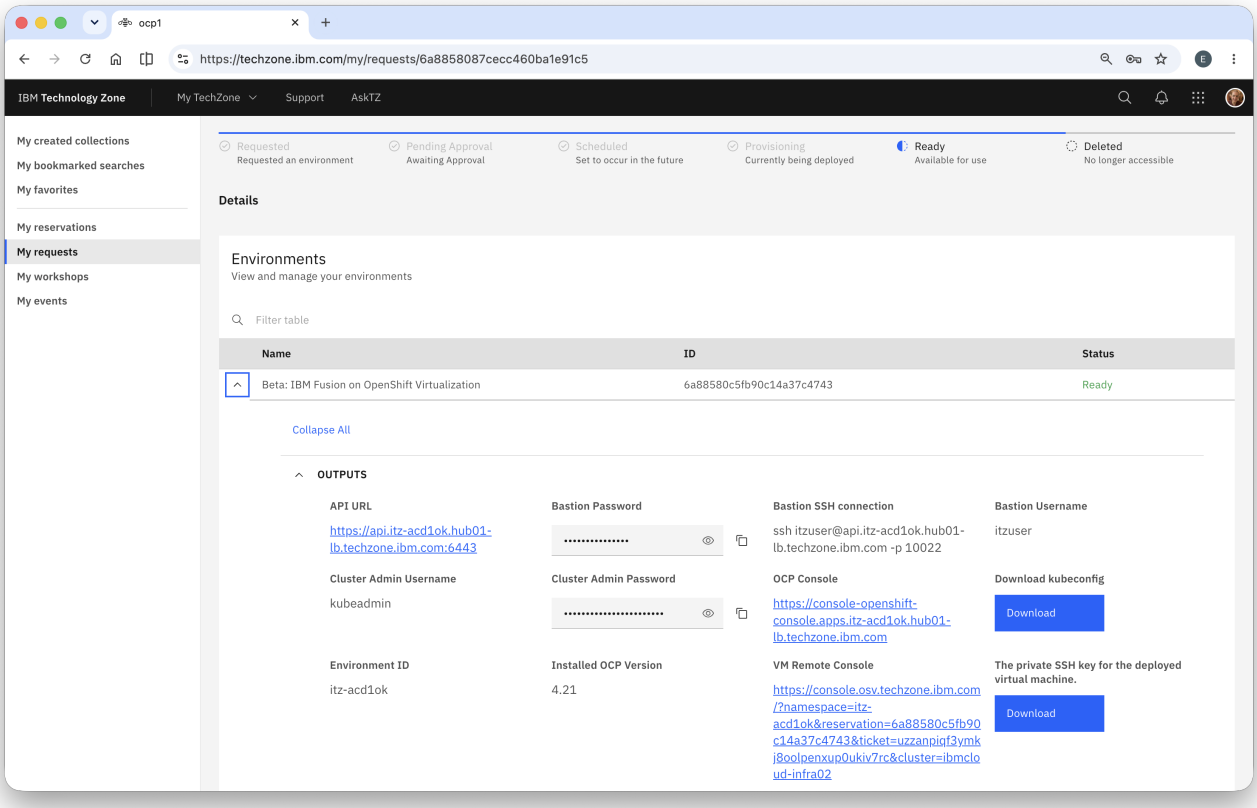
Once an environment reservation is successful you can start using it. If you are a first time then follow the steps below to get started.

Gather up some details

Start on the My requests page, locate the reservation you want to use, and make sure it is in the green "Reservation - Ready" state.

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Open the reservation page by clicking Open this reservation on the reservation tile (or the blue "eye" icon if you are in a list view instead of a tiled view.) Now expand the environment view by clicking the down-pointing arrow next to you environment name.



Before you continue we advise your to collects the various parameters of your environment such as passwords and access key file into a folder on your local desktop system. Create a folder which will be your work folder and collect the following artifacts into it:

- Download the kubeconfig file and save it
- Download the private SSH key and save it
- Create a text file and save the following parameters to it:
  - 24-character environment ID
  - Bastion user `itzuser` and the password from the hidden field
  - OCP cluster user `kubeadmin` and the password from the hidden field

For detailed instructions on connecting to the Web Console, SSH-ing to the Bastion, installing `oc` locally, and VM IP address references, see Accessing the Environment.

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Accessing Your TechZone Lab Environment

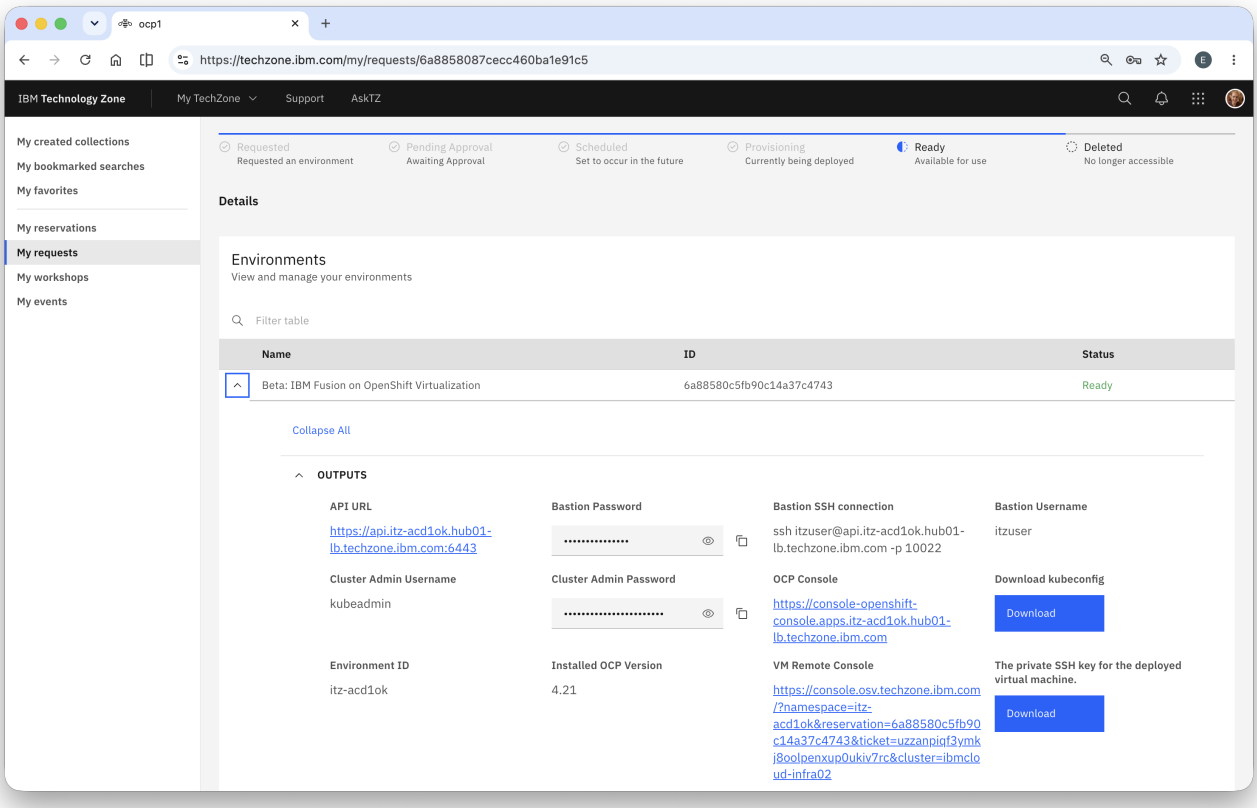
Introduction

Once your TechZone reservation is in the **Ready** state, you have three ways to interact with your OpenShift cluster. Most learners use a combination — the Web Console for UI-driven tasks and either the Bastion or a local `oc` install for CLI work.

Note If you have not yet provisioned a TechZone environment, complete the TechZone Quick Start first.

Gathering Your Credentials

All endpoints and credentials are found in the **Outputs** section of your reservation page. Navigate to My requests, open your reservation, and expand the environment row to reveal the Outputs panel shown below.



Before proceeding, collect the following into a working folder on your local machine:

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- Download the **kubeconfig** file — saves as `conf_kubeconfig_download.conf`
- Download the **private SSH key** — saves as `vm_ssh_key.vm` (check your Downloads folder)
- Note or copy the following values from the hidden fields (click the eye icon to reveal):
  - **Cluster Admin Password** (username is `kubeadmin`)
  - **Bastion Password** (username is `itzuser` — used for `sudo` on the Bastion, not SSH login)
- Note the **Bastion SSH connection** command — it is pre-built with the correct hostname and port
- Note the **API URL** — used for `oc login`

The following table describes every field in the Outputs panel:

Field	Description
API URL	OpenShift API endpoint used for `oc login` and kubeconfig
OCP Console	OpenShift Web Console URL — open directly in your browser
Cluster Admin Username	`kubeadmin`
Cluster Admin Password	Hidden — click the eye or copy icon
Bastion SSH connection	Pre-built SSH command — copy and paste directly into your terminal
Bastion Username	`itzuser`
Bastion Password	Hidden — required only for `sudo` on the Bastion, not for SSH login
Download kubeconfig	Download — saves as `conf_kubeconfig_download.conf`
The private SSH key	Download — required to authenticate SSH to the Bastion
VM Remote Console	Browser-based VNC access to the underlying cluster VMs
Environment ID	Your unique 24-character environment identifier
Installed OCP Version	The OpenShift version running in your cluster

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Important The Bastion host uses **key-based SSH authentication only** on port 10022. Password authentication is disabled on that port. You must download the private SSH key before attempting to connect.

Choosing an Access Method

Method	Best for	Requires
OpenShift Web Console	UI-driven tasks, monitoring, Software Catalog	Browser only
Bastion host (SSH)	CLI tasks — `oc` is pre-installed, no local setup needed	SSH client + private key
Local `oc` on your laptop	CLI tasks without connecting to the Bastion	`oc` installed + kubeconfig

Option 1: OpenShift Web Console

The OpenShift Web Console runs entirely in your browser — no software installation required.

1. From your TechZone reservation page, copy the **OCP Console** URL.
2. Open it in your browser.
3. On the login page, select the **kube:admin** identity provider.

   Tip Select **kube:admin** rather than IBMid to ensure you log in with full `cluster-admin` privileges.

4. Enter `kubeadmin` as the username and paste the **Cluster Admin Password** from your reservation outputs.
5. Upon successful authentication the OpenShift Administrator Dashboard is displayed.

Tip When your browser offers to save the password, accept — you will be logging back in frequently throughout the labs.

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Option 2: Bastion Host via SSH

The Bastion host has `oc`, `kubectl`, `helm`, and `curl` pre-installed. Connecting via SSH is the quickest way to run CLI commands without any local setup.

Step 1: Download the private SSH key

On your TechZone reservation page, click Download next to **The private SSH key for the deployed virtual machine**. The file saves as `vm_ssh_key.vm` (check your Downloads folder).

Step 2: Set key permissions

SSH requires the private key file to be readable only by your user.

macOS / Linux:

```
chmod 400 ~/Downloads/vm_ssh_key.vm
```

Windows (PowerShell):

```
icacls "$env:USERPROFILE\Downloads\vm_ssh_key.vm" /inheritance:r /grant:r
"$env:USERNAME:(R)"
```

Step 3: Connect via SSH

Copy the **Bastion SSH connection** command from your reservation page — it is pre-populated with the correct hostname and port. Add the `-i` flag pointing to your downloaded key:

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10022
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p
10022
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10022.

Step 4: Log in to OpenShift from the Bastion

Once connected to the Bastion, log in to your OpenShift cluster using the **API URL** and **Cluster Admin**

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Password from your reservation:

```
oc login -u kubeadmin -p <cluster-admin-password> https://api.<env-
id>.techzone.ibm.com:6443
```

Verify connectivity:

```
oc whoami
oc get nodes
```

`oc whoami` should return `kubeadmin` and `oc get nodes` should list all cluster nodes in `Ready` state.

Option 3: Local `oc` CLI on Your Laptop

If you prefer to run `oc` commands directly from your laptop without SSH-ing to the Bastion, install the `oc` CLI and use the downloaded kubeconfig.

Step 1: Install the `oc` CLI

macOS:

```
# Using Homebrew
brew install openshift-cli

# Verify
oc version
```

Alternatively, download the binary from the OCP Web Console: click the Help icon (?) in the top-right masthead and select Help › Command Line Tools, then download the macOS binary and move it to `/usr/local/bin/`.

Linux:

```
# Download from the OCP Web Console (Help > Command Line Tools), or use the mirror:
curl -LO https://mirror.openshift.com/pub/openshift-v4/clients/ocp/stable/openshift-
client-linux.tar.gz
tar -xvf openshift-client-linux.tar.gz
sudo mv oc /usr/local/bin/
oc version
```

Windows:

Download the `oc` binary from the OCP Web Console (Help icon → Help › Command Line Tools → **Windows**). Extract the zip and move `oc.exe` to a folder on your `PATH` (e.g. `C:\tools\`).

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Alternatively, install via Chocolatey:

```
choco install openshift-cli
oc version
```

Step 2: Download the kubeconfig

On your TechZone reservation page, click Download next to **Download kubeconfig**. The file saves as `conf_kubeconfig_download.conf`.

Step 3: Connect using the kubeconfig

macOS / Linux:

```
export KUBECONFIG=~/Downloads/conf_kubeconfig_download.conf
oc whoami
oc get nodes
```

Windows (PowerShell):

```
$env:KUBECONFIG = "$env:USERPROFILE\Downloads\conf_kubeconfig_download.conf"
oc whoami
oc get nodes
```

A successful connection returns `system:admin` from `oc whoami`.

Tip

To make the kubeconfig permanent across terminal sessions, merge it into your default `~/.kube/config`:

```
# macOS / Linux
KUBECONFIG=~/.kube/config:~/Downloads/conf_kubeconfig_download.conf \
  oc config view --flatten > /tmp/merged && mv /tmp/merged ~/.kube/config
```

SSH to IBM Storage Scale node

Option 1: Bastion node

To SSH from the Bastion directly into the IBM Storage Scale node:

```
ssh itzuser@10.10.10.202 □
```

□ Verify this IP against the VM Remote Console page before use.

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Option 2: Directly from your workstation

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10023.

VM IP Address Reference

The following IP addresses are assigned to VMs in the standard TechZone Fusion environment. Your configuration may include additional worker nodes — treat this as a representative reference:

VM	IP Address
`itz-*-bastion-1`	`10.10.10.201`
`itz-*-scale-*`	`10.10.10.202`
`itz-*-master-1`	`10.10.10.11`
`itz-*-master-2`	`10.10.10.12`
`itz-*-master-3`	`10.10.10.13`
`itz-*-infra-1`	`10.10.10.2`
`itz-*-infra-2`	`10.10.10.3`
`itz-*-infra-3`	`10.10.10.4`
`itz-*-worker-1`	`10.10.10.21`

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VM	IP Address
`itz-*-worker-2`	`10.10.10.22`
`itz-*-worker-3`	`10.10.10.23`

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IBM Fusion Operator Installation & Console Overview

Introduction

Although your Red Hat OpenShift cluster is provisioned on IBM Technology Zone, IBM Fusion has not yet been deployed. In this guide, you will install the IBM Fusion Operator from the Software Catalog, accept the product license, and explore the core administration interface.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Installing the IBM Fusion Operator

The IBM Operator Catalog and entitlement secrets are pre-configured on your TechZone OpenShift cluster.

Install via OpenShift Web Console

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. In the left navigation menu, go to Ecosystem › Software Catalog.
3. In the search field, type `Fusion` and select the **IBM Storage Fusion** operator tile.
4. Click Install.
5. Leave all installation options as default:
   - **Update channel:** Stable channel (e.g., `v2.x`)
   - **Installation mode:** All namespaces on the cluster (default)
   - **Installed Namespace:** `ibm-spectrum-fusion-ns` (or Operator-recommended namespace)
   - **Approval Strategy:** Automatic
6. Click Install at the bottom of the form.
7. Wait until the Operator status shows **Succeeded** / **Installed operator - ready for use**.

Optional: Install via OpenShift CLI (`oc`)

You can also verify or initiate operator installation from the Bastion terminal:

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```
oc get csv -n ibm-spectrum-fusion-ns
```

Connecting to the IBM Fusion Management Console

IBM Fusion is container-native and integrates directly into the OpenShift console.

1. From the top masthead of the OpenShift Web Console, click the **Application Menu** icon (the 9-square grid).
2. Click **IBM Storage Fusion** from the dropdown menu. A new browser tab opens to the Fusion interface.
3. If prompted with the OpenShift login page, select **kube:admin** and enter your cluster credentials.
4. On first launch, review and select **I have read and accept the license agreement**, then click Continue.

> **Note** Accepting the license initializes the core `SpectrumFusion` Custom Resource (CR) in the `ibm-spectrum-fusion-ns` namespace.

5. You are redirected to the **Welcome to IBM Fusion** dashboard.

Navigating the IBM Fusion Dashboard

The IBM Fusion interface provides a unified view of data management services:

- **Services:** View installed services and install optional capabilities (Fusion Data Foundation, Backup & Restore, Data Cataloging).
- **Storage:** Manage local storage clusters, StorageClasses, and remote IBM Storage Scale file systems.
- **Backup & Restore:** Configure backup locations (S3), retention policies, and application protection.
- **Events:** View Kubernetes-native events with severity filtering (Info, Warning, Critical).
- **Settings:** Review cluster networking, Call Home diagnostics, and cluster interconnect status.
- **Help & Support:** Access the question-mark icon in the masthead to download support logs or link to official IBM documentation.

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Configuring Fusion Data Foundation (FDF)

Introduction

In this guide you will enable the Data Foundation service in the IBM Fusion Console and configure the local Ceph storage cluster using the OpenShift Web Console wizard.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Enabling the Data Foundation Service

Note This guide covers the **Local** configuration mode, the standard mode for single-cluster labs. For an overview of all FDF configuration modes (Local, Dynamic, External, and Provider), see Persistent Storage: Architecture & Concepts.

1. In the IBM Fusion Console, navigate to **Services** in the left navigation menu.
2. Locate the **Data Foundation** service tile, click the tile, and then click Install.
3. On the **Install service** screen, select the **Local** tile and click Install.

   Note Selecting **Local** automatically installs the Local Storage Operator (LSO) — no separate LSO installation step is required.

4. Wait for the service status to show **Healthy**.

Configuring the Ceph Storage Cluster

Once the Data Foundation service is enabled, configure the local Ceph storage cluster through the OpenShift Web Console wizard launched from Fusion:

Step 1: Launch Storage Cluster Creation

1. In the IBM Fusion Console, navigate to Storage › Data Foundation (or **Local storage**).
2. Click Get Started (or Configure Data Foundation / Create Storage Cluster).

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3. This redirects to the **Create StorageSystem** wizard in the OpenShift Web Console.

Step 2: Backing Storage

1. On the **Backing storage** page, select **Create a new StorageClass using local storage devices**.
2. Click Next.

Step 3: Advanced Settings

1. On the **Advanced Settings** page, configure the default StorageClass options:
   - Leave **Use Ceph RBD as the default StorageClass** checked. This sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass after installation.
   - If this cluster will host virtual machines, also check **Set default StorageClass for virtualization**. This marks the RBD virtualization StorageClass as the default for KubeVirt VM disks after installation.
2. Leave all other options at their defaults and click Next.

Note Both default StorageClass options are configured here in the wizard. No additional manual annotation steps are required after installation.

Step 4: Create Local Volume Set

1. Enter `ibm-spectrum-fusion` as the **LocalVolumeSet name**.
2. Select **Disks on selected nodes** and choose the dedicated infra nodes (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`).
3. Set **Disk Type** to **SSD/NVMe**.
4. Click the `^` to reveal the **Advanced** section, and enter "100" in the minimum drive size field; this will exclude the 50GiB drives.
5. Click Next, then click Yes to confirm creation of the LocalVolumeSet.

Step 5: Capacity and Nodes

1. Verify the **Available raw capacity** populated from the discovered disks.
2. Select a **resource profile** — **Balanced** is recommended for lab environments.
3. Click Next.

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Step 6: Security and Network

1. Keep the default encryption settings unless KMS integration (Vault or Thales CipherTrust) is required.
2. Keep the default network option (**Default (OVN)**) unless using Multus.
3. Click Next.

Note If you are configuraing storage for Hosted Control Plane (HCP) then select the "Host" radio button.

Step 7: Data Protection

1. If this cluster will be configured for Regional Disaster Recovery, select **Prepare cluster for disaster recovery (Regional-DR only)**.
2. Otherwise leave unchecked and click Next.

Step 8: Review and Create

1. Review the configuration summary. Click Back to correct any settings.
2. Click Create StorageSystem.

Note Storage cluster initialization takes approximately 10–15 minutes. Wait until the cluster status reports **Ready / Healthy** under Storage › Data Foundation in the OpenShift console before proceeding.

Step 9: Verify StorageClasses and Ceph Cluster Health

From the OpenShift Web Console:

1. Navigate to Storage › StorageClasses and confirm the standard FDF storage classes are listed.
2. Navigate to Storage › Data Foundation and confirm the storage cluster status shows **Ready / Healthy**.
3. Navigate to Ecosystem › Installed Operators › OpenShift Data Foundation and confirm the operator status shows **Succeeded**.

From the CLI:

```
# Check provisioned StorageClasses
oc get sc

# Verify Ceph cluster status
```


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```
oc get storagecluster -n openshift-storage

# Confirm all ODF pods are Running or Completed
oc get pods -n openshift-storage
```

The following standard StorageClasses should now be available:

- `ocs-storagecluster-ceph-rbd` (Block storage for containers & VMs)
- `ocs-storagecluster-cephfs` (Shared file storage / RWX)
- `openshift-storage.noobaa.io` (Object Bucket Claims / S3)

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OpenShift Virtualization with IBM Fusion: Architecture & Overview

Introduction

OpenShift Virtualization (OCP-Virt / OSV) enables enterprises to run and manage legacy virtual machine (VM) workloads alongside modern containerized microservices on a single, unified Red Hat OpenShift platform.

When combined with **IBM Fusion**, VMs benefit from enterprise-grade persistent block storage (Fusion Data Foundation Ceph RBD), rapid VM live migration, snapshotting, and disaster recovery.

Key Capabilities

- **Unified Platform:** Run containers and VMs side-by-side on the same OpenShift cluster, managed through a single control plane.
- **Live Migration:** Move running VMs between worker nodes with zero downtime using KubeVirt's live migration engine — no shared storage or hypervisor-level clustering required.
- **Fusion Storage Integration:** VMs consume Ceph RBD block volumes provisioned by Fusion Data Foundation, providing enterprise snapshotting, cloning, and disaster recovery for VM disks.
- **virtctl CLI:** The `virtctl` command-line tool extends `oc` / `kubectl` with VM-specific operations including console access, start/stop, and live migration.

Architecture

OpenShift Virtualization is built on the **KubeVirt** project. Each VM runs as a standard Kubernetes pod (a `VirtualMachineInstance`) containing a `libvirt` / QEMU process. This means VMs are scheduled, monitored, and managed using the same Kubernetes constructs as containers — namespaces, resource quotas, network policies, and storage classes all apply.

Prerequisites

Before deploying VMs, the following operators must be installed from Ecosystem › Software Catalog and configured on the cluster. Full step-by-step instructions are in Install OCP-Virt.

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Operator	Required	Purpose
IBM Fusion	Yes	IBM Fusion base operator to enable IBM Fusion Console
IBM Fusion Data Foundation	Yes	IBM Fusion Data Foundation data services to provide volumes to virtual machines
IBM Fusion Backup & Restore Hub	Yes	IBM Fusion BnR to provide virtual machine backup and restore capabilities
OpenShift Virtualization	Yes	Provides the KubeVirt engine and the `HyperConverged` cluster resource
MetalLB	Optional	Bare-metal LoadBalancer for exposing VM services externally
Kubernetes NMState	Optional	Node-level network configuration for secondary interfaces (bridges, bonds, VLANs)
Migration Toolkit for Virtualization (MTV)	Optional	Migrates VMs from external sources (OVA/OVF) into OpenShift Virtualization

Important If you have not deployed the first three (3) operators above, please navigate back to the following chapters:

- IBM Fusion Base Operator installation
- IBM Fusion Data Foundation installation
- IBM Fusion Backup & Restore Overview

Software Versions

The following software versions were used to develop and validate the OpenShift Virtualization lab:

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Component	Version
IBM Fusion	2.14.x
Red Hat OpenShift Container Platform	4.21.33
Red Hat OpenShift Virtualization	4.21.17
IBM Fusion	2.14
Migration Toolkit for Virtualization	2.12.9
Kubernetes NMState	4.21.0-202609301335
MetalLB	4.21.0-202609291522

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Installing OpenShift Virtualization

Introduction

In this guide you will install the four operators required to support all OpenShift Virtualization lab exercises, initialize the HyperConverged cluster resource, and set up the shared lab prerequisites (projects, SSH key, `virtctl` CLI).

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Install Required Operators from Software Catalog

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. Navigate to Ecosystem › Software Catalog.
3. Install each of the following operators with default settings (default namespace, automatic approval) by searching for the name and clicking Install:

Operator	Purpose
OpenShift Virtualization	KubeVirt engine and `HyperConverged` cluster resource
Migration Toolkit for Virtualization (MTV)	Migrate VMs from external sources (OVA/OVF) into OSV
MetalLB	Bare-metal LoadBalancer for exposing VM services externally
Kubernetes NMState	Node-level network configuration (bridges, bonds, VLANs)

Initialize the HyperConverged Cluster

1. Navigate to Ecosystem › Installed Operators › OpenShift Virtualization Operator.

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2. Click the **HyperConverged** tab and select Create HyperConverged.
3. Scroll down to the `vmStateStorageClass` field and enter `ocs-storagecluster-cephfs`.

> **Note** `vmStateStorageClass` configures the storage class used to persist the virtual Trusted Platform Module (vTPM) state across VM restarts and live migrations. Required for Windows 2022 VMs.

4. Accept all other defaults and click Create.
5. Wait for the status to show **Available**. When a **Web console update is available** banner appears, click Refresh web console.

Verify from the CLI:

```
oc get hyperconverged -n openshift-cnv
```

The `PHASE` column should show `Available` before proceeding.

Initialize the MTV ForkliftController

1. Navigate to Ecosystem › Installed Operators › Migration Toolkit for Virtualization Operator.
2. Click the **ForkliftController** tab and select Create ForkliftController.
3. Leave the name as `forklift-controller` and click Create.
4. Wait for the status to show **Running**. Refresh the web console when prompted — a new **Migration** menu item will appear in the left navigation.

Verify from the CLI:

```
oc get forkliftcontroller -n openshift-mtv
```

The `STATUS` column should show `Running` before proceeding.

Initialize the NMState Instance

1. Navigate to Ecosystem › Installed Operators › Kubernetes NMState Operator.
2. Click Create instance in the **NMState** tile.
3. Accept all defaults and click Create.
4. Refresh the web console when prompted — **NodeNetworkConfigurationPolicy** and **NodeNetworkState** entries will appear under the **Networking** menu.

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Verify from the CLI:

```
oc get nmstate
```

The `AVAILABLE` column should show `True` before proceeding.

Red Hat OpenShift Virtualization Lab Setup

Create Three Projects

Create three projects that are used across the lab exercises:

```
oc new-project vmlab
oc new-project vmbackup
oc new-project vmnetwork
```

Or from the OpenShift Web Console: navigate to Home › Projects, click Create Project, and create each project in turn.

Create SSH Key for Virtual Machines

You will use an SSH key pair to connect to Linux VMs via `virtctl`. Generate the key pair on the Bastion host:

1. SSH to the Bastion host (see Accessing the Environment).
2. Generate a key pair, accepting all defaults:

   ```
   ssh-keygen -t rsa -b 2048
   ```
3. In the OpenShift Web Console, navigate to Virtualization › Overview › Settings › User.
4. Expand **Manage SSH keys**. Select project `vmlab` and click Not configured under **Authorized SSH key**.
5. Select Add new, then return to your Bastion terminal and print the public key:

   ```
   cat /home/itzuser/.ssh/id_rsa.pub
   ```
6. Copy the output and paste it into the **SSH key** field. Enter `bastion-ssh` as the **Secret Name** and click Save.

Download the `virtctl` CLI

The `virtctl` command-line tool extends `oc` with VM-specific operations such as console access, start/stop, and live migration. Download it directly from your cluster's web console to ensure version

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compatibility with your environment.

1. In the OpenShift Web Console, click the Help icon (?) in the top-right masthead and select **Command Line Tools**.
2. Scroll to the **virtctl – KubeVirt command line interface** section.
3. Right-click the **Download virtctl for Linux for x86_64** link and select **Copy Link Address**.
4. On the Bastion host, download and install `virtctl` using the copied URL:

```
curl -L -O <paste-download-url-here>
tar xzvf virtctl.tar.gz
chmod u+x virtctl
sudo mv virtctl /usr/local/bin/
virtctl version
```


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OpenShift Hosted Control Planes (HCP) with IBM Fusion: Overview

Introduction

Hosted Control Planes (HCP) decouple the OpenShift control plane (API server, etcd, controllers) from worker nodes by running the control plane components as lightweight container pods inside a centralized management cluster. This decoupled architecture enables greater flexibility, scalability, and resource efficiency compared to traditional OpenShift clusters where each cluster requires its own dedicated control plane nodes.

The IBM Fusion system can host multiple OpenShift clusters simultaneously. Clusters can be created and deleted on demand, allowing environments to be tailored to specific workload needs.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Key Benefits

- **Cost efficiency** — No dedicated control plane nodes per hosted cluster; fewer bare metal nodes required overall. Compute nodes for hosted clusters can be provisioned as KubeVirt VMs or bare metal servers depending on workload needs.
- **Cluster on demand** — Hosted clusters can be created when needed and deleted when no longer in use. Resources are automatically reclaimed and made available for new deployments.
- **Shared storage** — IBM Fusion Data Foundation in Provider mode supplies a shared storage backend for all hosted cluster worker VMs and etcd, eliminating per-cluster storage overhead.

HCP Architecture

In an HCP deployment:

- **Management Cluster** — Hosts the HyperShift operator, Red Hat multicluster engine for Kubernetes (MCE), MetalLB for load balancing, and the control plane pods for all hosted clusters. IBM Fusion Data Foundation runs in Provider mode to supply storage for local workloads, OpenShift Virtualization, and all hosted clusters.

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- **Hosted Clusters** — Consist solely of worker nodes, running as KubeVirt virtual machines on the management cluster, that connect back to their containerized control plane pods.

Lab Components

The HCP lab environment includes the following node types on the management cluster:

Node Type	Role
3 Control Plane nodes	Run the OpenShift management cluster control plane.
3 Infra nodes (`itz-*-infra-*`)	Serve a dual role: host the 50 GiB LVM volumes used as fast local storage for hosted cluster etcd pods, and provide the 500 GiB local block devices for Fusion Data Foundation in Provider mode.
3 Worker nodes (`itz-*-worker-*`)	Host user workloads and KubeVirt virtual machines (the worker nodes of hosted clusters).

Note Infrastructure nodes host only infrastructure components and are not counted toward the total number of subscriptions required to run the environment.

LVM Storage for Hosted Cluster etcd

Hosted cluster etcd pods have strict latency requirements. To meet these, a fast dedicated storage class is required for etcd on each control-plane node. In this lab, **LVM Storage** is used to create a local storage class backed by the dedicated 50 GiB disks on the infra nodes.

IBM Fusion 2.10 and later includes a `lvm-config` ConfigMap mechanism that automates the installation and configuration of the Red Hat LVM Operator and its associated `LVMCluster` custom resource after the Fusion Operator is installed.

Fusion Data Foundation in Provider Mode

FDF Provider mode runs a local storage cluster that acts as the central storage provider for multiple consumer clusters. The architecture has two components:

- **Provider** — The FDF storage cluster, built on a minimum of three worker nodes using local bare metal volumes. External clusters connect through a client operator. FDF replicates data across nodes for

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resilience and fault tolerance.

- **Client** — A managed CSI deployment on each hosted cluster that connects to the provider to deliver a local storage class for hosted cluster workloads.

From FDF 4.19 onward, Provider and Local deployments share a unified configuration. If you plan to use Provider mode, ensure that Multus is **not** enabled during deployment.

Key Prerequisites and Dependencies

Deploying Hosted Control Planes requires the following capabilities on the management cluster, all of which are covered step by step in Provisioning Hosted Control Plane Clusters:

1. **LVM Storage** — Provides fast local storage for hosted cluster etcd pods on infra nodes.
2. **Fusion Data Foundation (Provider mode)** — Provides shared block storage for KubeVirt VM disks and workloads.
3. **OpenShift Virtualization** — Runs hosted cluster worker nodes as KubeVirt virtual machines.
4. **MetalLB** — Provides Layer 2 load balancing and external IP assignment for hosted cluster API endpoints and OAuth routes.
5. **multicluster engine for Kubernetes (MCE)** — Provides cluster lifecycle management and the HyperShift operator engine for creating and managing hosted clusters.

Software Versions

The following software versions were used to develop and validate the HCP lab:

Component	Version
Red Hat OpenShift Container Platform	4.21.33
Red Hat OpenShift Virtualization	4.21.17
Red Hat multicluster engine for Kubernetes	2.17.3
Red Hat LVM Storage	4.21.1
IBM Fusion	2.14

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Component	Version
Migration Toolkit for Virtualization	2.12.9
Kubernetes NMState	4.21.0-202609301335
MetalLB	4.21.0-202609291522

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Provisioning Hosted Control Plane Clusters on KubeVirt

Introduction

In this lab you will prepare the management cluster and deploy a tenant OpenShift cluster using Hosted Control Planes (HCP). Worker nodes for the hosted cluster run as KubeVirt virtual machines backed by IBM Fusion Data Foundation storage.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Prerequisites

The following must be completed before starting this lab. Each item is covered in its own guide — follow the links if you have not yet completed them.

- IBM Fusion Operator installed and license accepted
- OpenShift Virtualization Operator installed and HyperConverged initialized

Note OpenShift Virtualization is required because hosted cluster worker nodes run as KubeVirt virtual machines on the management cluster.

Lab Preparation

The following steps are unique to the HCP lab and are performed here for the first time.

Configure LVM Storage

LVM Storage provides fast local storage for hosted cluster etcd pods. Each infra node (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`) has multiple 50 GiB disks available for this purpose. The 500 GiB disk on each node is reserved for Fusion Data Foundation.

IBM Fusion automates LVM Operator installation and configuration using an `lvm-config` ConfigMap. You need to provide the infra node names and the stable PCI device path of one of the 50 GiB disks.

Important Use the stable PCI device path (under `/dev/disk/by-path/`) rather than a kernel

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device name like `/dev/vdb`. PCI paths are persistent across reboots; kernel device names are not guaranteed to be stable.

Step 1: Label the Infra Nodes

From the OpenShift Web Console:

1. Navigate to Compute › Nodes and search for `infra`.
2. Click on `itz-*-infra-1` to open the Node details page.
3. Select the **Details** tab and click the pencil icon next to **Labels**.
4. Enter `node-role.kubernetes.io/infra` and click Save.
5. Repeat steps 2–4 for `itz-*-infra-2` and `itz-*-infra-3`.

From the CLI:

```
for node in $(oc get nodes -o name | grep infra); do
  oc label $node node-role.kubernetes.io/infra=
done

# Verify the labels were applied
oc get nodes -l node-role.kubernetes.io/infra
```

All three infra nodes should appear in the output before proceeding.

Step 2: Discover the Stable Disk Path

1. In the OpenShift Web Console, navigate to Compute › Nodes and click on `itz-*-infra-1`.
2. Select the **Terminal** tab and enter `chroot /host` to access the node filesystem.
3. Run `lsblk` to view the disk layout. You should see output similar to the following:

```
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
vda    252:0    0   300G  0 disk
|-vda1 252:1    0     1M  0 part
|-vda2 252:2    0   127M  0 part
|-vda3 252:3    0   384M  0 part /boot
`-vda4 252:4    0 299.5G  0 part /sysroot ...
vdb    252:16   0    50G  0 disk
vdc    252:32   0    50G  0 disk
vdd    252:48   0    50G  0 disk
vde    252:64   0   500G  0 disk
```

The unpartitioned 50 GiB disks (`vdb`, `vdc`, `vdd`) are the LVM candidates. The 500 GiB disk (`vde`) is for

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FDF and must not be used here.

4. Get the stable PCI path for one of the 50 GiB disks (for example, `vdb`):

```
ls -l /dev/disk/by-path/pci* | grep vdb
```

Copy the full symlink target path (for example, `/dev/disk/by-path/pci-0000:08:00.0`).

5. Repeat steps 1–4 for `itz-*-infra-2` and `itz-*-infra-3` and note the path for each.

Note In the TechZone OSV environment the PCI path is typically the same across all three infra nodes. If they differ, list each path on its own line in the `drives` field of the ConfigMap.

Step 3: Create the `lvm-config` ConfigMap

1. Click the Quick create icon (the `+` icon in the masthead) and select **Import YAML**.
2. Copy the following ConfigMap, replacing `itz-` with your environment ID and the PCI path with the value you discovered in Step 2:

```
kind: ConfigMap
apiVersion: v1
metadata:
  name: lvm-config
  namespace: ibm-spectrum-fusion-ns
data:
  computeNodes: |
    - itz-******-infra-1
    - itz-******-infra-2
    - itz-******-infra-3
  drives: |
    - '/dev/disk/by-path/pci-0000:XX:00.0'
  nodeType: compute
```

3. Click Create.
4. Fusion will automatically install and configure the Red Hat LVM Operator. Wait for the LVM Operator to show **Succeeded** under Ecosystem › Installed Operators before continuing.

Verify from the CLI:

```
oc get csv -n openshift-storage | grep lvm
```

The `PHASE` column should show `Succeeded` before continuing.

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Configure Fusion Data Foundation in Provider Mode

FDF Provider mode supplies the central storage backend for KubeVirt VM disks and hosted cluster workloads. See Fusion Data Foundation in Provider Mode for a conceptual overview.

Install the Data Foundation Service

1. In the IBM Fusion Console, navigate to **Services**, locate the **Data Foundation** tile, and click Install.
2. Select the **Local** tile and click Install.

> **Note** Selecting **Local** automatically installs the Local Storage Operator (LSO). No separate LSO installation step is required.

3. Wait for the service status to show **Healthy**.

Configure the Storage Cluster

1. In the IBM Fusion Console, navigate to Storage › Data Foundation and click Get Started.
2. This redirects to the **Create StorageSystem** wizard in the OpenShift Web Console.

Backing storage page:

Select **Create a new StorageClass using local storage devices** and click Next.

Advanced Settings page:

1. Leave **Use Ceph RBD as the default StorageClass** checked.
2. Check **Set default StorageClass for virtualization**.
3. Click Next.

> **Note** Both default StorageClass options are configured here in the wizard. No additional manual annotation steps are required after installation.

Create Local Volume Set page:

1. Enter `ibm-spectrum-fusion` as the **LocalVolumeSet name**.
2. Select **Disks on selected nodes** and choose the three infra nodes (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`).
3. Set **Disk Type** to **SSD/NVMe**.
4. Expand **Advanced** and set **Minimum Disk Size** to `100` GB to exclude the 50 GiB LVM disks from FDF disk discovery.

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5. Click Next, then click Yes to confirm creation of the LocalVolumeSet.

Capacity and Nodes page:

1. Verify the available raw capacity reflects the 500 GiB disks on the infra nodes.
2. Select **Balanced** as the resource profile and click Next.

Security and network page:

Select **Host** network and click Next.

Data Protection page:

Leave unchecked unless this cluster is being configured for Regional Disaster Recovery, then click Next.

Review and create:

Review the configuration summary and click Create StorageSystem. Wait until the Storage Cluster status shows **Ready / Healthy** — this typically takes 10–15 minutes.

Verify from the CLI:

```
oc get storagecluster -n openshift-storage
```

The `PHASE` column should show `Ready` before continuing.

Install MetalLB

MetalLB provides Layer 2 load balancing and external IP assignment for hosted cluster API endpoints and OAuth routes.

1. Navigate to Ecosystem › Software Catalog, search for `MetalLB`, and select the **MetalLB Operator** tile.
2. Click Install, keep all defaults, and click Install.
3. Once installed, navigate to Ecosystem › Installed Operators › MetalLB Operator (project: `metallb-system`).
4. On the **MetalLB** tab, click Create instance. Accept all defaults and click Create.
5. Click the **IPAddressPool** tab → Create IPAddressPool:
    - **Name:** `techzone-ipaddresspool`
    - **Addresses:** `10.10.10.205-10.10.10.250`
    - Remove any extra address fields, then click Create.

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6. Click the **L2Advertisement** tab → Create L2Advertisement:
   - **Name:** `l2-adv-techzone`
   - **ipAddressPools:** `techzone-ipaddresspool`
   - Click Create.

Patch the ingress controller to allow wildcard routes required by HCP:

```
oc patch ingresscontroller -n openshift-ingress-operator default \
  --type=json \
  -p
'[{"op":"add","path":"/spec/routeAdmission","value":{"wildcardPolicy":"WildcardsAllowed"}}
]'
```

Install Red Hat multicluster engine for Kubernetes (MCE)

MCE provides cluster lifecycle management and the HyperShift operator engine for Hosted Control Planes.

1. Navigate to Ecosystem › Software Catalog, search for `multicluster engine for Kubernetes`, and select the operator tile.
2. Click Install, keep all defaults, and click Install.
3. Wait for the Create MultiClusterEngine button to become active, then click it.
4. Keep all defaults and click Create.
5. Wait for the `MultiClusterEngine` resource to show status **Available / Reconcile Complete**.
6. When the **Web console update is available** popup appears, click **Refresh web console**. After the reload, the **Fleet management** interface is available via Core Platform › Fleet Management.

Verify from the CLI:

```
oc get multiclusterengine
```

The `STATUS` column should show `Available` before continuing.

Download CLI Tools

Download the `virtctl` CLI

The `virtctl` command-line tool extends `oc` with VM-specific operations such as console access, start/stop, and live migration. Download it directly from your cluster's web console to ensure version

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compatibility with your environment.

1. In the OpenShift Web Console, click the Help icon (?) in the top-right masthead and select **Command Line Tools**.
2. Scroll to the **virtctl – KubeVirt command line interface** section.
3. Right-click the **Download virtctl for Linux for x86_64** link and select **Copy Link Address**.
4. On the Bastion host, download and install `virtctl` using the copied URL:

```
curl -L -O <paste-download-url-here>
tar xzvf virtctl.tar.gz
chmod u+x virtctl
sudo mv virtctl /usr/local/bin/
virtctl version
```

Download the `hcp` CLI

The `hcp` command-line tool is used to create and manage Hosted Control Plane clusters. Download it directly from your cluster's web console to ensure version compatibility with your environment.

1. In the OpenShift Web Console, click the Help icon (?) in the top-right masthead and select **Command Line Tools**.
2. Scroll to the **hcp – Hosted Control Plane Command Line Interface (CLI)** section.
3. Right-click the **Download hcp for Linux for x86_64** link and select **Copy Link Address**.
4. On the Bastion host, download and install `hcp` using the copied URL:

```
curl -L -O <paste-download-url-here>
tar xzvf hcp.tar.gz
chmod u+x hcp
sudo mv hcp /usr/local/bin/
hcp version
```

Generate an SSH Key for Hosted Clusters

On the Bastion host, generate an SSH keypair for accessing hosted cluster worker nodes:

```
ssh-keygen -t ed25519 -f /home/itzuser/.ssh/id_ed25519 -N ""
cat /home/itzuser/.ssh/id_ed25519.pub
```

Note the public key output — it will be required when creating the hosted cluster.

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Save the Cluster Pull Secret

The cluster pull secret authenticates to the Red Hat image registry when creating hosted clusters.

From the OpenShift Web Console:

1. Navigate to Workloads › Secrets and switch to the `openshift-config` project.

> **Note** Enable **Show default projects** in the project selector if `openshift-config` is not visible.

2. Click on the `pull-secret` secret, scroll to the **Data** section, and click the **Copy to clipboard** icon.
3. Paste the contents into a file on the Bastion (for example, `/home/itzuser/pull-secret.json`).

From the CLI:

```
oc get secret pull-secret -n openshift-config \
  -o jsonpath='{.data.\.dockerconfigjson}' | base64 -d \
  > /home/itzuser/pull-secret.json
```

Create a Hosted Cluster

Add Cloud Provider Credentials

1. Navigate to Core Platform › Fleet Management → **All Clusters**.
2. Go to Credentials › Add credential and select **Red Hat OpenShift Virtualization** as the credential type.
3. Provide a name and enter the contents of your saved pull secret.
4. Paste the SSH public key from `/home/itzuser/.ssh/id_rsa.pub`.
5. Click Add.

Create the Hosted Cluster

1. In Fleet Management, navigate to Infrastructure › Clusters and click Create cluster.
2. Select **Red Hat OpenShift Virtualization** as the infrastructure type, then select the **Hosted** tile.
3. On the **Cluster details** page, configure the following fields:

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Field	Value
Infrastructure provider credential	Select the credential created in the previous step
Cluster name	`demo-hcp-cluster`
Hosted cluster namespace	`clusters`
Cluster set	`default`
Release image	Select the latest OpenShift 4.21.x image
Etcd storage class	`lvms-hcp-etcd`

4. In the **Additional labels** field, add the following labels (press ， after each to apply as a tag):

```
isf.ibm.com/fusion-base=
isf.ibm.com/fusion-fdf=
isf.ibm.com/fusion-backup=
```

5. Click Next.

6. On the **Node pools** page, configure the node pool:

Field	Value
Node pool name	`demo-hcp-cluster-nodepool`
Node pool replicas	`3`
Cores	`8`
Memory (GiB)	`16`

7. Expand the **Root volume** section and configure:

Field	Value
Root volume storage class	`ocs-storagecluster-ceph-rbd-virtualization`

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Field	Value
Access mode	`ReadWriteMany`
Volume mode	`Block`

Note The root volume storage class configures the PVC backing each KubeVirt worker node VM disk. `ocs-storagecluster-ceph-rbd-virtualization` is the FDF storage class optimised for VM workloads. `ReadWriteMany` with `Block` volume mode is required for live migration of hosted cluster worker VMs.

8. Click Next through any remaining pages, then click Create cluster.
9. Monitor control plane pod creation on the management cluster:

```
oc get pods -n clusters-demo-hcp-cluster -w
```

10. Track hosted cluster and node pool readiness:

```
oc get hostedclusters -n clusters
oc get nodepools -n clusters
```

Accessing the Hosted Cluster

Once the hosted cluster reports `AVAILABLE: True`:

1. In Fleet Management, select the `demo-hcp-cluster` cluster and click Download kubeconfig.
2. Save the kubeconfig and connect to the hosted cluster:

```
export KUBECONFIG=~/demo-hcp-kubeconfig
oc get nodes
oc get clusterversion
```


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